

Final Project: You Are the Contractor

Week 10 — Tuesday Lecture

Today

- What the project is, and what makes it engineering
- Four turn-ins, two flight days, one 50-minute trip
- Permissions: airspace, property, people
- Project ideas, old and new — including a 3D model of your own group
- **Before you leave:** a partner and two candidate sites

The job

You have been hired to fly a drone mission for a client: a city, a contractor, BYU Physical Facilities.

1. Find a civil or construction use for drone data
2. Plan the project, then the flight
3. Fly it, process it, measure something
4. Present it, and write a one-page memo

Anyone can produce an orthomosaic. The engineering is saying who needs the data, why you flew it that way, and what it can and cannot be used for.

"We flew over a parking lot."

Is that a project? What is missing?

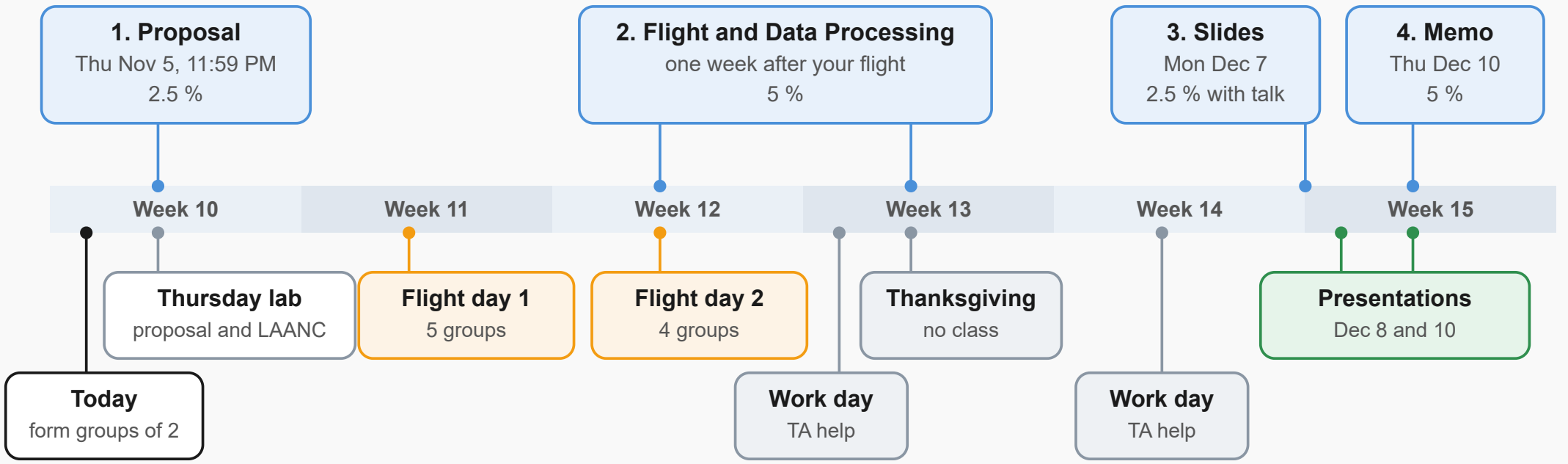
ANSWER A client, a question, and a product that answers it

Weak	Strong
"We flew over a parking lot."	"How much asphalt would it take to repave this lot, and what would it cost?"
No one needs the data	Physical Facilities budgets the work from it
A pretty 3D model	An area, a volume, a count, a slope, a map of damage

The question decides the drone, the altitude, the pattern — and the measurement.

Figure 1: The final project, week by week

Fall 2026. Learning Suite has the official dates; if they differ, Learning Suite wins.



Flights leave EB B122 at 8:00 or 9:00 and last 50 minutes. Final project work cannot be turned in late.

Groups and flight days

- **Groups of 2.** Same score for both, except the presentation: a student who does not present gets a zero
- **Thu Nov 12** (5 groups) or **Thu Nov 19** (4 groups), trips at **8:00–8:50** or **9:00–9:50**
- Meet at **EB B122**. The trip leaves on time — a partner who is late is left behind
- Sign up for a **drone on a trip** on the sign-up sheet; first come, first served
- A second flight on another day can be arranged with the TAs, if your project needs one
- No class on a flight day for the groups not flying

Three drones per trip

Drone	Use it for
Mini	Regular photos: orthomosaics, 3D models, areas, counts
Thermal	Regular photos plus heat: roofs, pavement, moisture
Multispectral	Plant health (NDVI) plus regular photos

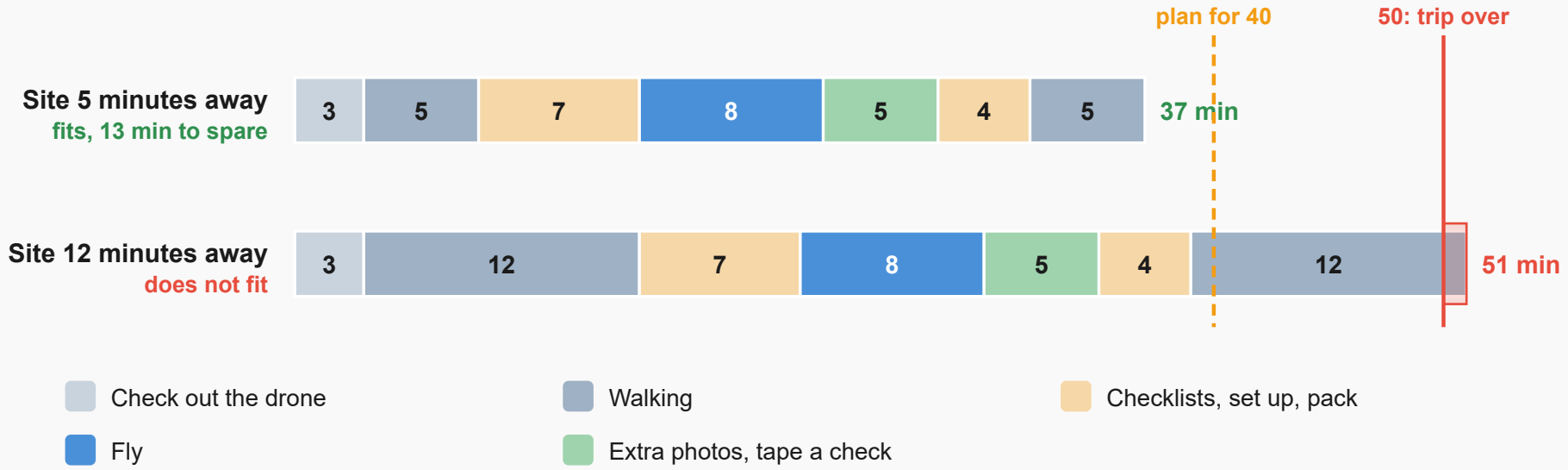
The thermal and multispectral slots go first. **If regular photos answer your question, plan on the Mini** — and name a backup drone in your proposal.

Your site is a 12-minute walk from EB.

The flight itself is 8 minutes. Does the trip fit in 50?

Figure 3: Does the trip fit in 50 minutes?

From EB B122 and back. The next trip needs the drone, so plan for 40 and keep 10 for wind, people and batteries.



Minutes are an example. Time your own walk, and take the flight time from your plan.

Permissions are your job

- **Airspace.** Most of campus is in Provo's **Class D**. Flying there needs **LAANC** authorization — Low Altitude Authorization and Notification Capability. A TA files it with you on Thursday, under their Part 107 certificate
- **Property.** Private property, active construction sites and some campus areas need the owner's OK, in writing. An email is fine
- **People and traffic.** Plan the flight so it never passes over people or moving cars

Every flight has a certified TA or instructor supervising.

LAANC approved for 200 ft. Can you fly over the Quad at noon?

ANSWER No — LAANC is airspace only

- LAANC answers one question: may you be in this controlled airspace, this high, at this time
- It says nothing about **people** under you (§107.39), the **owner** of the ground, or **moving vehicles**
- Fly the Quad at 7 a.m., or pick a site where no one walks under the grid

Three separate checks: airspace, property, people. The proposal names all three.

The proposal — due Thursday 11:59 PM

About one page, plus the flight-plan screenshot and Mission Log:

1. **Project and purpose** — client, problem, decision
2. **Site** — map, why it fits, hazards
3. **Airspace, permissions, safety** — with the LAANC request attached
4. **Time budget** — every minute, 50 or less
5. **Flight plan** — every setting with a reason
6. **Products and roles** — what you will measure, who flies, who watches

Every setting gets a "because"

Setting	Your value	Why
Drone, and a backup		
Altitude above ground		
Ground sample distance (GSD)		
Front and side overlap		
Gimbal angle		
Pattern: grid, double grid, orbit, ascending circle		
Flight time and batteries		

"110 ft, 80 % front" is a list. "110 ft, because a 0.5 in GSD shows the cracks" is a plan.

Two gates

A proposal whose flight is **not legal**, or **does not fit in 50 minutes**, comes back — whatever else it does well.

Fix it and resubmit before your flight day. **You are not cleared to fly until the TAs approve it.**



Project ideas

Use one, adapt one, or bring your own

Proven ideas from past semesters

Idea	Product	What went wrong before
Parking lot resurfacing	Area, stall count, cost	Missed a corner — run the grid past every edge
Pavement condition	Map of the worst failures	Trees modeled badly; fly low enough for a 0.5 in GSD
Court or park remodel	Orthomosaic, DSM for drainage	Smooth surface left holes — raise the overlap
Earthwork volume	Area × depth, or a DSM	Area came out as 100,000 square miles: wrong CRS
Room for more parking	Proposed stalls, dimensioned	Measure one real stall first to check the scale

Full notes on the Example Projects page.

New: quick projects near campus

Walkable from EB, one battery, about 30 minutes on site.

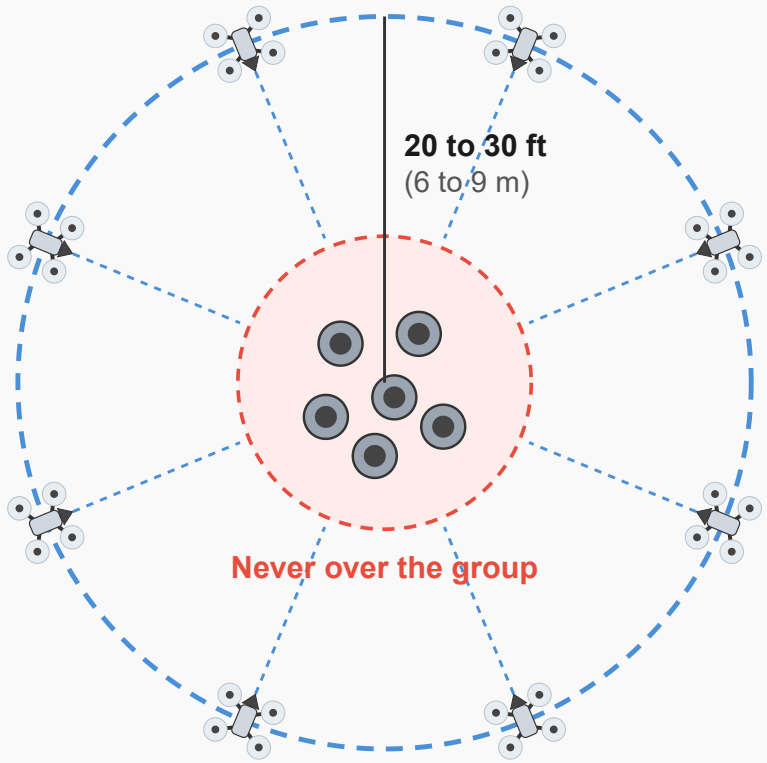
Idea	Drone	Pattern	Check one number
A 3D model of your group	Mini	Ascending circle	Tape one person's height
How tall is it? A light pole, flagpole, building corner	Mini	Ascending circle	Plans, or tape the shadow
Roof heat loss at 8 a.m.	Thermal	Grid	Thermometer on one surface
Is this ramp accessible?	Mini	Low grid + oblique line	A level on the ramp
Count the lot by computer	Mini	Grid, two days	Hand-count one row

Your proposal names the exact site. These are starting points.

Figure 2: A 3D model of your group

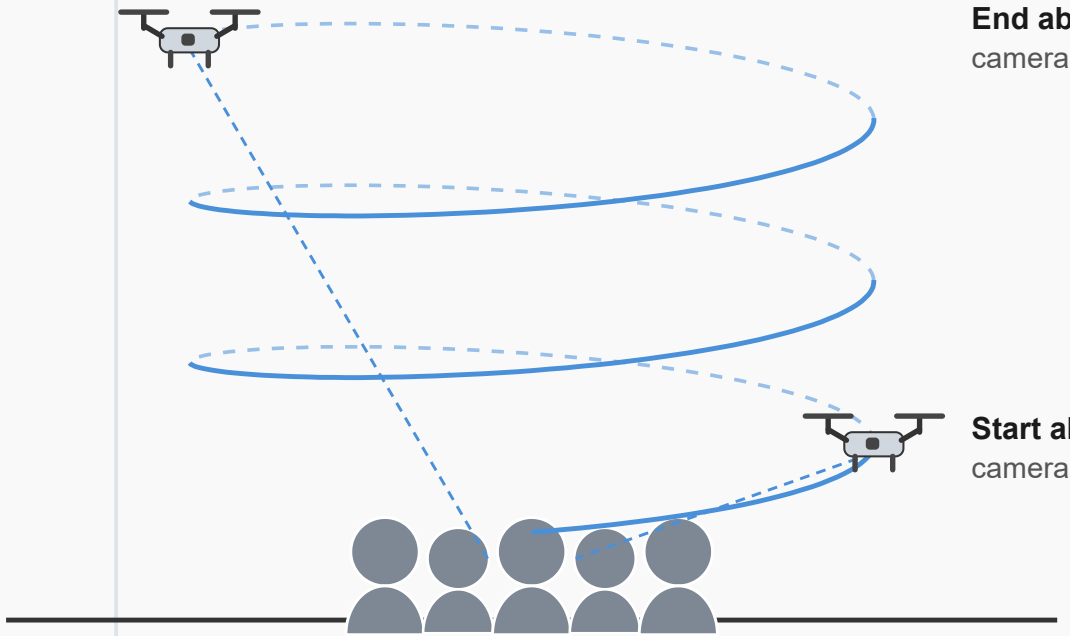
An ascending circle, the same pattern DJI's tower missions fly. Numbers are a starting point; tune them in your proposal.

From above



Camera pointed in the whole way round,
a photo every 10° or so

From the side



The group holds still for 3 to 5 minutes.
Anyone who moves becomes a blur or a ghost.

The group portrait: is that a flight over people?

The drone circles 25 ft out with the camera pointed in.

ANSWER Not if the circle stays outside the group

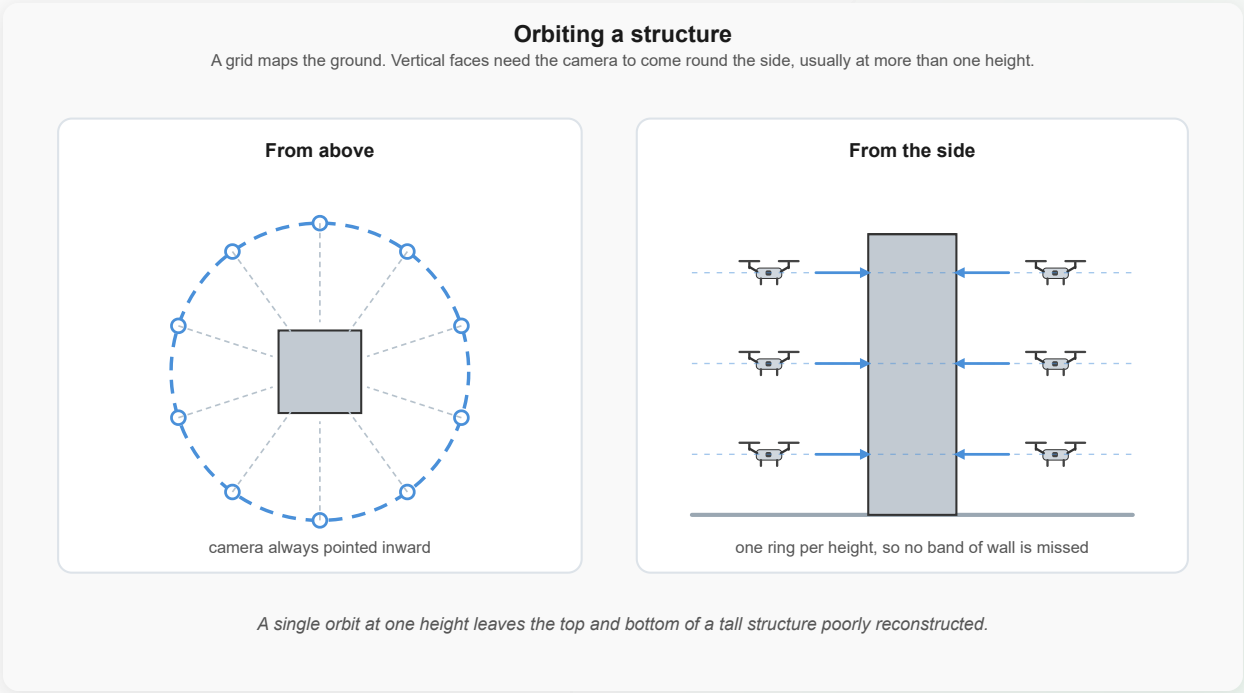
- §107.39: no flight **over** a person who is not directly taking part in the operation
- The drone circles **around** the group; it never passes over a head
- Anyone else who wanders inside the circle: pause the mission
- Pick a spot with no foot traffic — a lawn at 8 a.m., not the Quad at noon

You want the height of a light pole. Will a nadir grid do it?

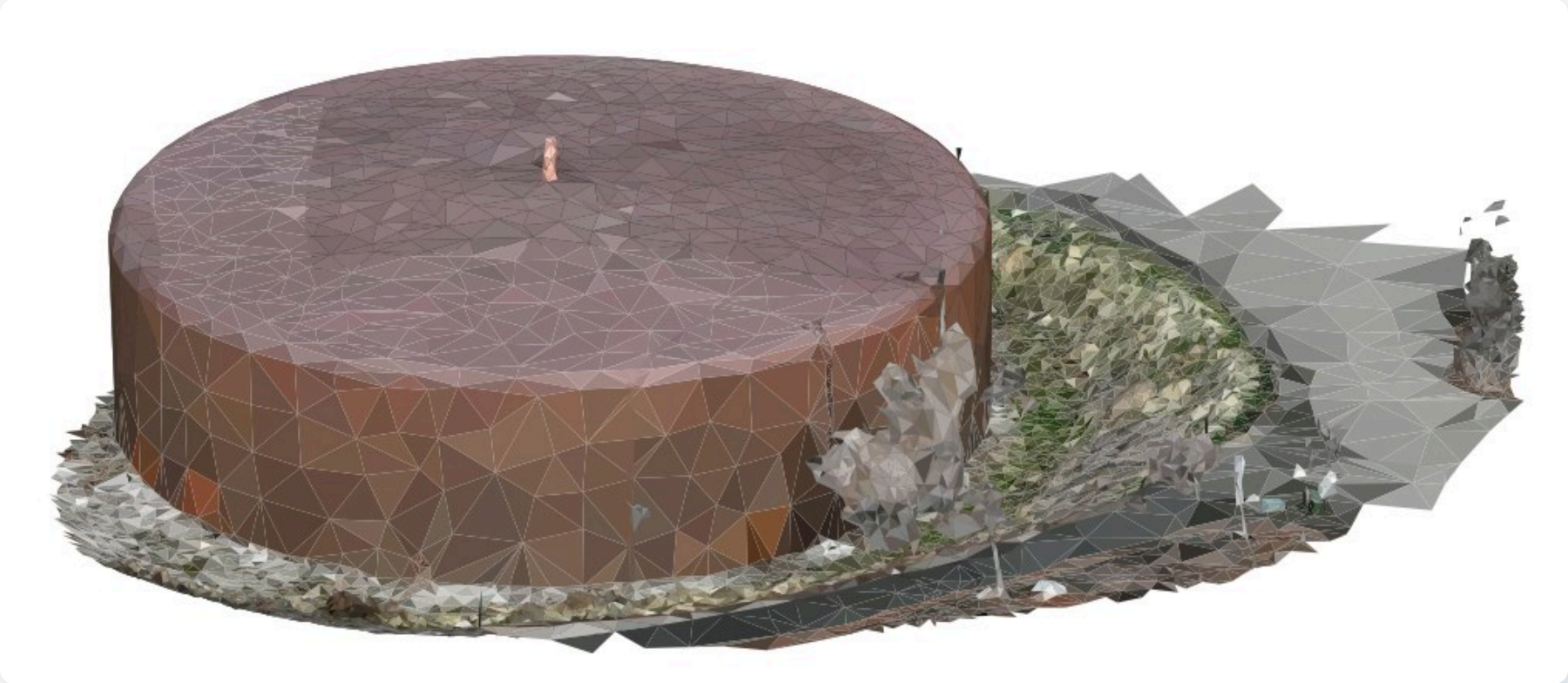
A nadir grid: camera straight down, lawnmower lines.

ANSWER No — straight down, a pole is a dot

- From above you see the top of the pole and almost none of its side
- A thin object gets too few matches; it comes out as a blob, or a web, or not at all
- Circle it with the camera tilted in, climbing as you go — the same plan as the group portrait



A class 3D object: the Rock Canyon water tank



Flown by a class in March 2026: a grid plus one line of oblique photos at −45°. The wall is there because the oblique line saw it.

Roof heat loss at 8 a.m.

- Cold November morning, before the sun reaches the roof: the building has heated all night, so a warm patch is heat leaking out — wet insulation, a gap, an unsealed vent
- Fly the Autel on a grid with about **90 %** front and side overlap; thermal photos have little texture
- **Power on 10 minutes early**, then run the shutter calibration (FFC). A cold camera read up to 30 °C too cold — Week 8's −8 °C parking lot
- Photo format **JPG + TIFF**: the `.irg` and `.TIFF` hold the temperatures, the colour JPG is only a picture
- **Limit to say in the memo:** apparent temperature, not true temperature — shiny metal and glass lie

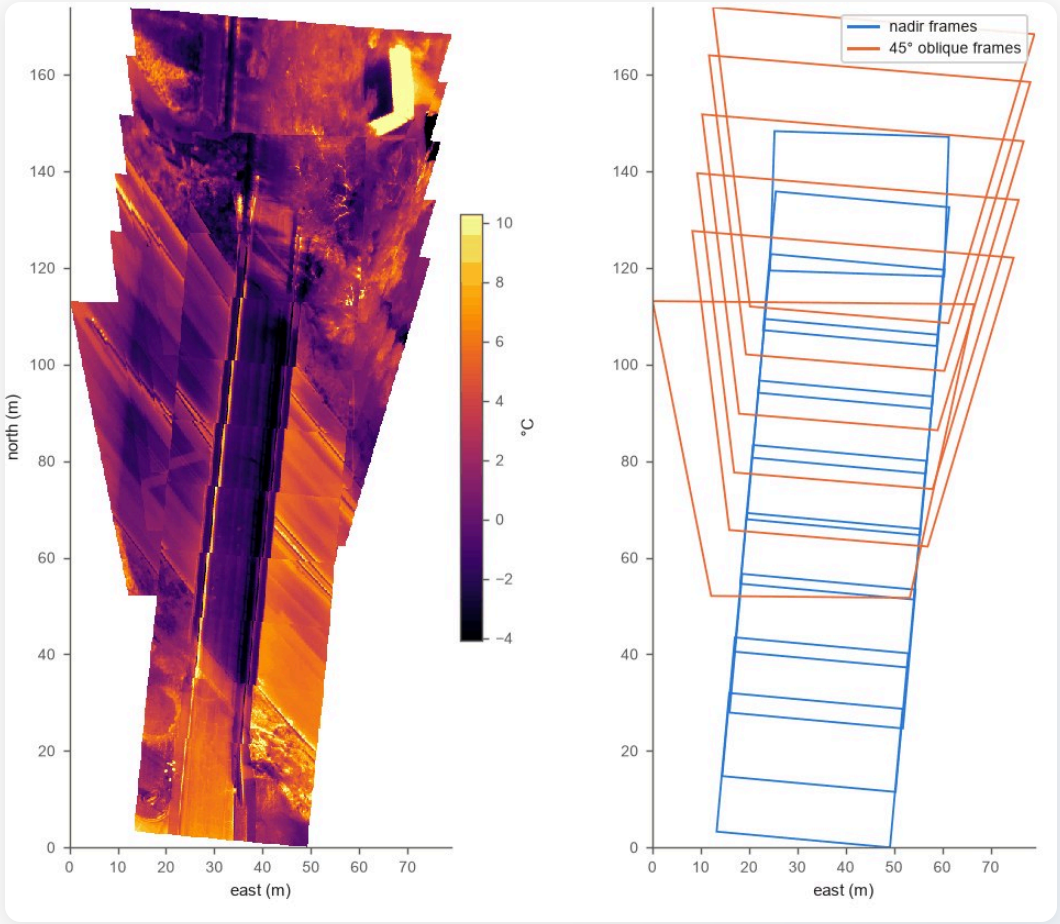
From frames to a temperature map: autel-thermal

The class's own Python tool, written for this camera.

```
pip install "autel-thermal[notebooks] @ git+https://github.com/gus-p-williams/autel-thermal"
autel-ortho roof_flight roof_out --mosaic
```

- Copy the whole flight folder off the SD card — `.irg`, `.TIFF`, `.jpg` and the `.mrk` GPS log — and rename nothing
- Converts every frame to °C (matches Autel's own software, bit for bit), places it on the map in UTM metres, and stitches `mosaic.tif`
- Leaves out frames with the cold-camera fault and says why in `ortho_report.csv`
- Open `mosaic.tif` in QGIS, or work through the four example notebooks — 04 finds the hot spots and their area in m²

What you get: a class flight



Carterville, Oct 2025, 16 frames at 61 m. Left: the mosaic in °C. Right: each frame's outline.

The tool puts every frame on flat ground at the takeoff height. Your roof is 6 m up and you fly at 40 m. What happens to the area of a warm patch?

ANSWER It comes out about 40 % too large

- The camera is 34 m above the roof, but the tool thinks 40 m, so every roof pixel is drawn 40 ÷ 34 = **1.18** times too long — and too wide
- Area: 1.18 × 1.18 ≈ **1.38**
- Fix: put the plane at the roof — `--ground-below-takeoff -6` (negative means above the takeoff point). Then measure only on the roof
- Check: one roof edge from the plans or a tape, measured again in QGIS

Is this ramp accessible?

- The ADA limit for a ramp is a slope of **1:12**, about **8.3 %**
- Fly a low grid along the ramp, plus one oblique line; make a DSM; read the rise over the run in QGIS
- Check it with a 2 ft level and a tape at two spots on the ramp
- A useful answer for Physical Facilities: which runs meet 1:12, which do not

Count the lot by computer

Fly a lot, make the orthomosaic, and let a Python image-recognition model count the cars and trucks.

- **The model:** YOLO11 nano, the smallest and fastest version, in its oriented bounding box (OBB) variant. The weights download automatically the first time it runs
- **Trained on DOTA v1**, aerial images labelled with 15 classes: plane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, large vehicle, small vehicle, helicopter, roundabout, soccer ball field and swimming pool
- **Oriented** means each box can rotate to match how the car is parked, instead of being a box aligned to the image edges
- Fly it twice — two days, or 8:05 and 8:40 — and the project becomes an **occupancy study**
- The check: count one row by hand. How many did the model miss, and how many did it invent?

A starting script: resample first

```
import cv2
from ultralytics import YOLO                          # pip install ultralytics
model = YOLO("yolo11n-obb.pt")
img = cv2.imread("lot.tif")                           # 8-bit RGB export
scale = 0.02 / 0.12                                   # native GSD / target GSD
img = cv2.resize(img, None, fx=scale, fy=scale, interpolation=cv2.INTER_AREA)
r = model.predict(img, imgsz=max(img.shape[:2]), conf=0.3, max_det=3000)[0]
names = [r.names[int(c)] for c in r.obb.cls]
print({n: names.count(n) for n in set(names)})   # small vehicle, large vehicle
cv2.imwrite("lot_boxes.jpg", r.plot())                # every box drawn: check one row
```

- **Pixel size:** DOTA cars are mostly 20–60 px long. Squeeze a 20,000 px orthomosaic to 1,024 px and a car is 11 × 5 px; leave it at 2 cm and it is 225 px. Both miss. At **10–15 cm/px** a car is 30–50 px
- `max_det` defaults to 300: a lot with more cars is cut off at 300, silently
- **Export first:** JPEG stops at 65,535 px per side, and OpenCV may choke on BigTIFF, 4-band or float GeoTIFFs. Save an 8-bit RGB copy

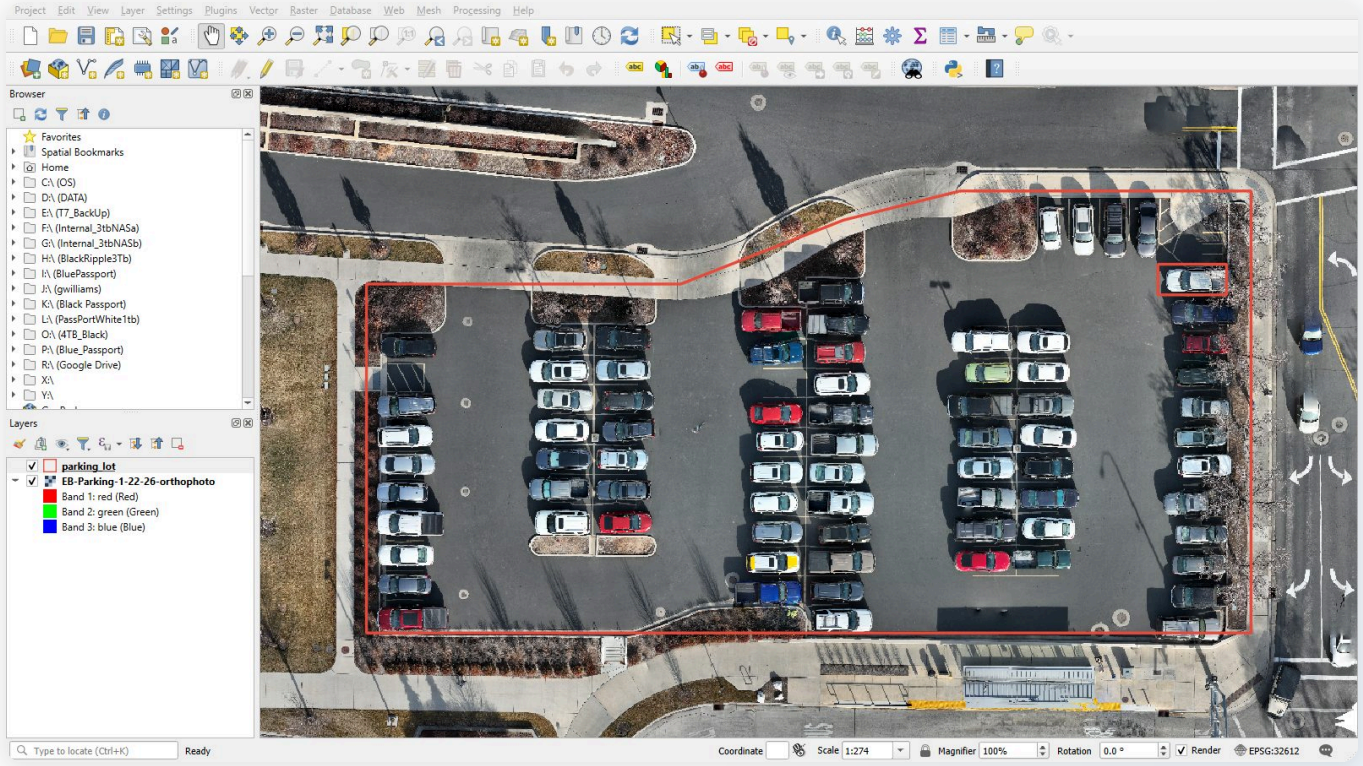
What good work looks like

- The project answers a question someone would actually ask
- The flight plan is made **before** flying, and every setting has a reason
- The flight runs the way a client would expect from a contractor
- The results are clear, and you say plainly what went wrong and what the data cannot support

Check one number. Compare one drone measurement with something measured another way: a taped length, a stall count, a dimension from a plan.

The memo: one page, six headings

Scope and purpose · Flight plan · Flight execution · Data summary and results · Limitations · Conclusion



From the Example Technical Memo: the EB east lot traced in QGIS, 28,000 ft², and one stall taped at 19.5 ft against 19.4 ft on the map.

What can the client *not* decide from your data?

Take the parking-lot memo: area 28,000 ft², cost about $98,000.

ANSWER Say it before they ask

- **Parked cars** hid a third of the pavement: no crack map, so no crack-sealing option
- **Shadows and trees** blur one edge: the area could be off by a few hundred ft²
- The **cost per square foot** is a typical value, not a quote: good for a budget, not a bid

Each limitation says what went wrong, how much it matters, and what would fix it.

The presentation: 12 minutes + 5 of questions

- Both partners present. Slides due **Mon Dec 7**; present **Dec 8 or Dec 10**
- **Lead with the answer** in the first minute, then show how you got it
- Show your site, your products, your measurement, and what went wrong
- Skip the history of drones and the software tour

Questions to expect: Why that altitude and overlap? What would you change? How sure are you of that number? What can the client not decide?

Thursday: proposal lab

- Bring your partner, your idea and **two** candidate sites
- Build the flight plan in the class planner; fill in the Mission Log
- File the **LAANC** request with a TA; email any property owner **that day**
- Proposal due **Thu Nov 5, 11:59 PM**, one PDF per group, with the KMZ and permissions



Before you leave today

Find a partner · pick two sites · sign up for a drone on a trip